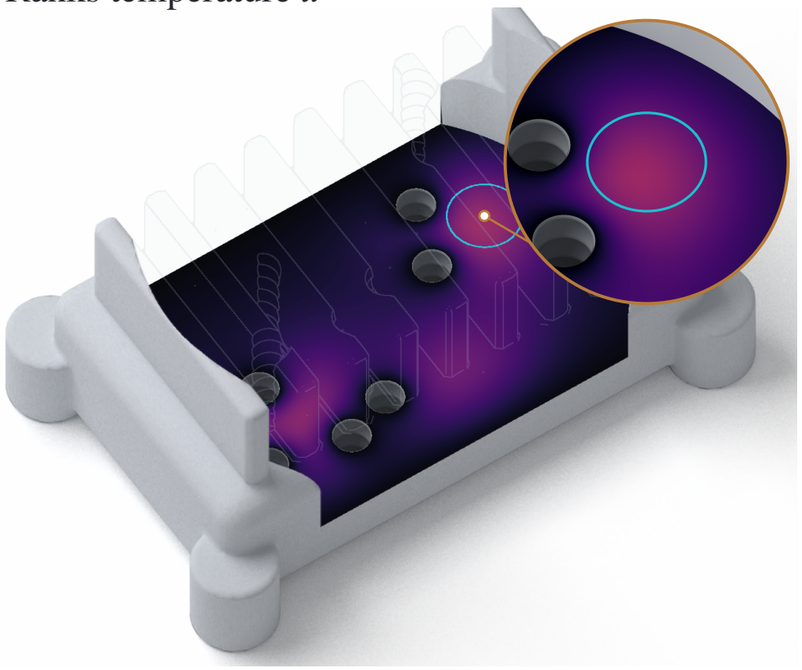
Hottest-point placement

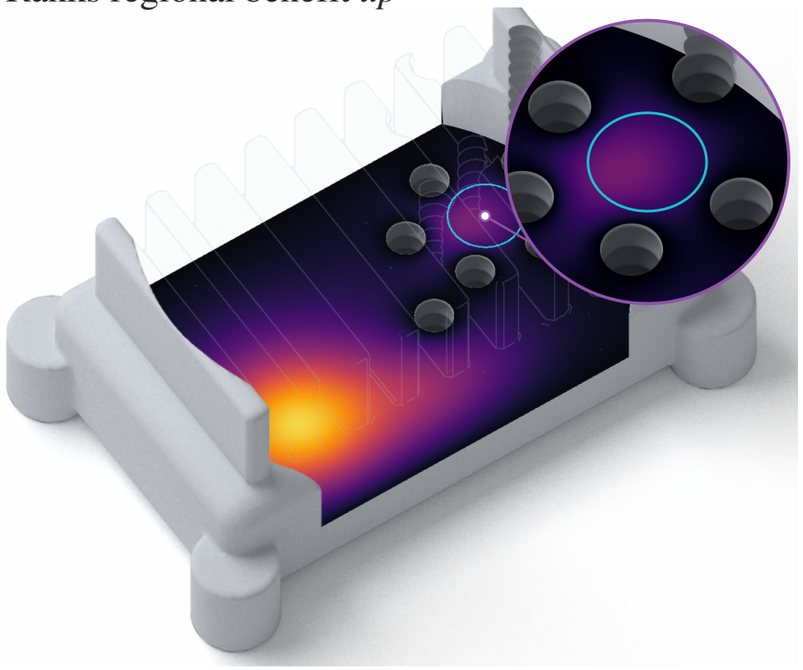
Ranks temperature *u*



33.9% lower regional temperature

TWoS placement

Ranks regional benefit *up*



54.1% lower regional temperature

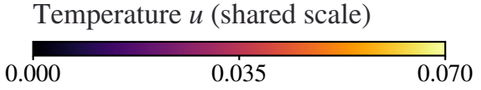
Multigrid placement

Ranks *up* on a volumetric grid



58.0% lower regional temperature

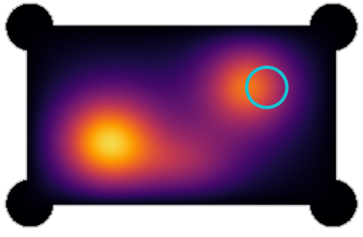
First right-sensor seed. Cyan: sensor; faint shells: cut-away bars.



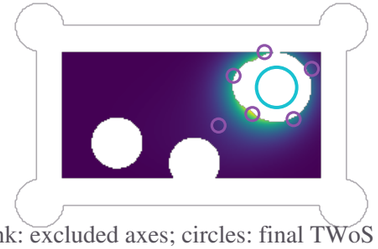
Why the placements differ

Temperature *u* × target influence *p* gives placement priority.

Original temperature *u*



Admissible priority *up*



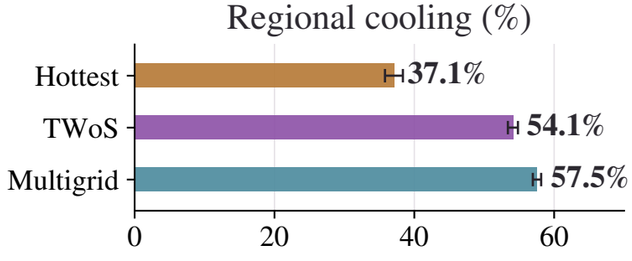


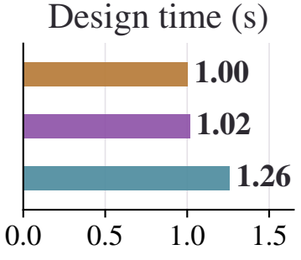
Blank: excluded axes; circles: final TWoS axes

Ideal cold walls; point-score heuristic. Display cuts are excluded from the PDE.

The measured advantage

Both sensor positions; five paired seeds; six channels per design.





+17.0 points vs. hottest-point placement at similar cost.

Multigrid cools more; TWoS is faster here. Full study: 50 designs.